



Dyslipidemia

(Hyperlipidemia)

By [Michael H. Davidson](#), MD, FACC, FNLA, University of Chicago Medicine, Pritzker School of Medicine;

[Marie Altenburg](#), MD, The University of Chicago

Reviewed/Revised May 2025 | Modified Mar 2026

Dyslipidemia is a high level of cholesterol and/or triglycerides or a low level of high-density lipoprotein (HDL) cholesterol.

- Lifestyle, genetics, disorders (such as low thyroid hormone levels or kidney disease), medications, or a combination can contribute.
- [Atherosclerosis](#) can result, causing angina, heart attacks, strokes, and peripheral arterial disease.
- Doctors measure levels of triglycerides and the various types of cholesterol in the blood.
- Exercise, dietary changes, and drugs can be effective.

(See also [Overview of Cholesterol and Lipid Disorders](#).)

Important fats (lipids) in the blood are:

- Cholesterol
- Triglycerides

Cholesterol is an essential component of cell membranes, of brain and nerve cells, and of bile, which helps the body absorb fats and fat-soluble vitamins. The body uses cholesterol to make vitamin D and various hormones, such as estrogen, testosterone, and cortisol. The body can produce all the cholesterol that it needs, but it also obtains cholesterol from food.

Triglycerides, which are contained in fat cells, can be broken down, then used to provide energy for the body's metabolic processes, including growth. Triglycerides are produced in the intestine and liver from smaller fats called [fatty acids](#). Some types of fatty acids are made by the body, but others must be obtained from food.

[Lipoproteins](#) are particles of proteins and other substances. They carry fats, such as cholesterol and triglycerides, which cannot circulate freely in the blood by themselves.

There are different types of lipoproteins (see table [Major Lipoproteins: Lipid Carriers](#)), including:

- Chylomicrons

- High-density lipoproteins (HDL)
- Low-density lipoproteins (LDL)
- Very low density lipoproteins (VLDL)

Levels of lipoproteins and therefore lipids, particularly low-density lipoprotein (LDL) cholesterol, increase slightly as people age. Levels are normally slightly higher in men than in women, but levels increase in women after menopause. The increase in levels of lipoproteins that occurs with age can result in dyslipidemia.

The risk of developing [atherosclerosis](#) increases as the total cholesterol level (which includes LDL cholesterol, HDL cholesterol, and VLDL cholesterol) increases, even if the level is not high enough to be considered dyslipidemia. Atherosclerosis can affect the arteries that supply blood to the heart (causing [coronary artery disease](#)), those that supply blood to the brain (causing [stroke](#)), and those that supply the rest of the body (causing [peripheral artery disease](#)). Therefore, having a high total cholesterol level also increases the risk of having a [heart attack](#) or [stroke](#).

Having a low total cholesterol level is generally considered better than having a high one.

Although there is no natural cutoff between normal and abnormal cholesterol levels, for adults, a total cholesterol level of less than 200 milligrams per deciliter of blood (mg/dL [< 5.1 mmol/L]) is desirable. And many people benefit from keeping the lipid level even lower. In parts of the world (such as China and Japan) where the average cholesterol level is 150 mg/dL (3.8 mmol/L), coronary artery disease is less common than it is in countries such as the United States. The risk of a heart attack more than doubles when the total cholesterol level approaches 300 mg/dL (7.7 mmol/L).

The total cholesterol level is only a general guide to the risk of atherosclerosis. Levels of the components of total cholesterol—particularly LDL and HDL cholesterol—are more important. A high level of LDL (bad) cholesterol increases the risk. Current evidence suggests that both low and high HDL levels can be associated with increased cardiovascular risk. Experts consider an LDL cholesterol level of less than 100 mg/dL (2.6 mmol/L) desirable.

Whether high triglyceride levels increase the risk of a heart attack or stroke is uncertain. Triglyceride levels higher than 150 mg/dL (1.7 mmol/L) are considered abnormal, but high levels do not appear to increase risk for everyone. For people with high triglyceride levels, the risk of heart attack or stroke is increased if they also have a low HDL cholesterol level, [diabetes](#), [chronic kidney disease](#), or many close relatives who have had [atherosclerosis](#) (family history).

Lipoprotein (a) is a combination of LDL with an additional protein attached to it. Levels higher than about 30 mg/dL (or 75 nmol/L) are associated with increased risk of atherosclerosis. High levels are inherited. Lipoprotein (a) is not affected by diet or most lipid-lowering drugs. It usually only needs to be measured once.

TABLE

Desirable Lipid Levels in Adults*

Lipid	Desirable Level
Total cholesterol	Less than 200 mg/dL (5.1 mmol/L)†
Low-density lipoprotein (LDL) cholesterol	Less than 100 mg/dL (2.6 mmol/L)
High-density lipoprotein (HDL) cholesterol	More than 60 mg/dL (1.6 mmol/L)
Triglycerides	Less than 150 mg/dL (1.7 mmol/L)
* These levels are a suggested guideline. Some people may need treatment even if their lipids are at these levels if they have risk factors or diseases such as coronary artery disease or stroke. † mg/dL = milligrams per deciliter of blood; mmol/L = millimoles per liter of blood.	

Causes of Dyslipidemia

Factors that cause dyslipidemia are categorized into

- Primary causes: Genetic (hereditary) causes
- Secondary causes: Lifestyle and other causes

Both primary and secondary causes contribute to dyslipidemia in varying degrees. For example, a person with a hereditary hyperlipidemia may have even higher lipid levels if the person also has secondary causes of hyperlipidemia.

Primary (hereditary) dyslipidemia

Primary causes involve gene mutations that cause the body to produce too much LDL cholesterol or triglycerides or to fail to remove those substances. Some causes involve underproduction or excessive removal of HDL cholesterol. Primary causes tend to be inherited and thus to run in families. Some of the genetic causes of dyslipidemia are discussed here and [elsewhere in The Manual](#).

Cholesterol and triglyceride levels are highest in people with primary dyslipidemias, which interfere with the body's metabolism and elimination of lipids. People can also inherit a tendency for HDL cholesterol to be unusually low.

Consequences of primary dyslipidemias can include premature [atherosclerosis](#) (in men 55 years old or younger, in women 60 years old or younger), which can lead to [angina](#) or [heart attacks](#). [Peripheral artery disease](#) is also a consequence, often causing decreased blood flow to the legs, with pain during walking ([claudication](#)). [Stroke](#) is another possible consequence. Very high triglyceride levels can cause [pancreatitis](#).

In people who have a genetic disorder that causes high triglyceride levels (such as familial hypertriglyceridemia or familial combined hyperlipidemia), certain disorders and substances can increase triglycerides to extremely high levels. Examples of disorders include poorly controlled [diabetes](#) and [chronic kidney disease](#). Examples of substances include excessive alcohol consumption and use of certain medications, such as estrogens (taken by mouth), that increase triglyceride levels.

Symptoms can include fat-containing bumps (eruptive xanthomas) in the skin on the front of the legs and back of the arms, enlargement of the spleen and liver, abdominal pain, and decreased sensitivity to touch due to nerve damage. Primary dyslipidemias can cause [pancreatitis](#), which is occasionally fatal.

Treatment of primary dyslipidemia begins with a healthy diet and exercise. Losing weight and not drinking alcohol can also help. Lipid-lowering medications may be effective.

Familial combined hyperlipidemia

In familial combined hyperlipidemia, the levels of cholesterol, triglycerides, or both may be high. This disorder affects about 1 to 2% of people. The lipid levels typically become abnormal after age 30 years but sometimes at a younger age, especially in people whose diet is very high in fat, or who have obesity or the [metabolic syndrome](#).

Treatment of familial combined hyperlipidemia involves limiting intake of saturated fat, cholesterol, and sugar as well as exercising and, when applicable, losing weight. Many people with this disorder need to take lipid-lowering medications.

Familial dysbetalipoproteinemia

In familial dysbetalipoproteinemia, levels of very low density lipoprotein (VLDL) cholesterol, total cholesterol, and triglycerides are high. These levels are high because an unusual form of VLDL accumulates in the blood. Fatty deposits (eruptive xanthomas) may form in the skin over the elbows and knees and in the palms, where they can cause reddish or yellowish bumps. This uncommon disorder results in the early development of severe atherosclerosis. By middle age, [atherosclerosis](#) often produces blockages in the coronary and peripheral arteries.

Treatment of familial dysbetalipoproteinemia involves achieving and maintaining recommended body weight and limiting intake of saturated fats and carbohydrates. A lipid-lowering medication is usually needed. With treatment, lipid levels can be improved, the progression of atherosclerosis may be slowed, and the fatty deposits in the skin may become smaller or disappear.

Familial hypercholesterolemia

In familial hypercholesterolemia, the total cholesterol level is high. People may have inherited one abnormal gene or they may have inherited two abnormal genes, one from each parent. People who have two abnormal genes (homozygotes) are more severely affected than people who have only one abnormal gene (heterozygotes). About 1 in 200 people are heterozygotes, and 1 in 250,000 to 1 in 1 million people are homozygotes.

Affected people may have fatty deposits in the tendons at the heels, knees, elbows, and fingers. Rarely, xanthomas appear by age 10. Familial hypercholesterolemia can result in rapidly progressive atherosclerosis and early death due to [coronary artery disease](#). Children with two abnormal genes may have a heart attack or angina by age 20, and men with one abnormal gene often develop coronary artery disease between ages 30 and 50. Women with one abnormal gene are also at increased risk, but the risk usually starts about 10 years later than in men. People who smoke or have high blood pressure, diabetes, or obesity may develop atherosclerosis at even younger ages.

Treatment of familial hypercholesterolemia begins with following a diet that is low in saturated fats. When applicable, losing weight, stopping smoking, and increasing physical activity are advised. One or more lipid-lowering medications are usually needed. Some people require [apheresis](#), a method of filtering the blood to lower LDL cholesterol levels. Some people with homozygous familial hypercholesterolemia may benefit from [liver transplantation](#). Early diagnosis and treatment can decrease the increased risk of heart attack and stroke.

Familial hypertriglyceridemia

In familial hypertriglyceridemia, triglyceride levels are high. This disorder affects about 0.2 to 1% of people. In some families affected by this disorder, [atherosclerosis](#) tends to develop at a young age, but in others, it does not.

Losing weight, if necessary, and limiting alcohol and carbohydrate consumption often lower triglyceride levels to normal. If these measures are ineffective, use of a lipid-lowering medication can help. For people who also have [diabetes](#), good control of the diabetes is important.

Hypoalphalipoproteinemia

In hypoalphalipoproteinemia, the HDL cholesterol level is low. Many different genetic abnormalities can cause the low HDL level. Since medications that raise HDL cholesterol do not decrease the risk of atherosclerosis, hypoalphalipoproteinemia is treated by lowering LDL cholesterol.

Familial chylomicronemia syndrome

Familial chylomicronemia syndrome (formerly called lipoprotein lipase deficiency) is a rare disorder caused by the lack of certain proteins needed for the removal of triglyceride-containing particles. In this disorder, the body cannot remove chylomicrons from the bloodstream, resulting in very high triglyceride levels. Without treatment, levels are often considerably higher than 1,000 mg/dL (11 mmol/L) and can cause [pancreatitis](#).

Symptoms appear during childhood and young adulthood. They include

- Recurring bouts of abdominal pain
- Enlargement of the liver and spleen
- Pinkish yellow bumps in the skin on the elbows, knees, buttocks, back, front of the legs, and back of the arms

These bumps, called eruptive xanthomas, are deposits of fat. Eating fats worsens symptoms. Although this disorder does not lead to [atherosclerosis](#), it can cause [pancreatitis](#), which is occasionally fatal.

People who have these disorders must strictly limit the amount of all types of fat—saturated, unsaturated, and polyunsaturated—in their diet. People may need to take vitamin supplements to make up for nutrients missing from their diets. There are some therapies in development for treatment of lipoprotein lipase deficiency and apolipoprotein CII deficiency.

Secondary dyslipidemia

Secondary causes contribute to many cases of dyslipidemia.

The **most important secondary cause** of dyslipidemia is:

- A sedentary lifestyle with excessive dietary intake of total calories, saturated fat, cholesterol, and trans fats (see sidebar [Types of Fat](#))

Some **other common secondary causes** include the following:

- Having [diabetes mellitus](#)
- Consuming large amounts of alcohol
- Having [chronic kidney disease](#)
- Having [hypothyroidism](#)
- Having [primary biliary cirrhosis](#)
- Using certain medications

Some people are more sensitive to the effects of diet than others, but most people are affected to some degree. One person can eat large amounts of animal fat, and the total cholesterol level does not rise above desirable levels. Another person can follow a strict low-fat diet, and the total cholesterol does not fall below a high level. This difference seems to be mostly genetically determined. A person's genetic makeup influences the rate at which the body makes, uses, and disposes of these fats. Also, body type does not always predict levels of cholesterol. Some people who have obesity have low cholesterol levels, and some people who have a thin body type have high levels. Eating excess calories can result in high triglyceride levels, as can consuming large amounts of alcohol.

Did You Know...

- Body type does not predict cholesterol levels. Some people with obesity have low cholesterol levels, and some people with thin body types have high cholesterol levels.

Some disorders cause lipid levels to increase. [Diabetes](#) that is poorly controlled or [chronic kidney disease](#) can cause total cholesterol levels or triglyceride levels to increase. Some liver disorders (particularly [primary biliary cirrhosis](#)) and an underactive thyroid gland ([hypothyroidism](#)) can cause the total cholesterol level to increase.

Use of medications such as estrogens (taken by mouth), oral contraceptives, corticosteroids, retinoids, thiazide diuretics (to some extent), [cyclosporine](#), [tacrolimus](#), and antiviral drugs used to treat [human immunodeficiency virus](#) (HIV) infection can cause cholesterol and/or triglyceride levels to increase.

Cigarette smoking, HIV infection, poorly controlled diabetes, or kidney disorders (such as nephrotic syndrome) may contribute to a low HDL cholesterol level. Medications such as beta-blockers and anabolic steroids can lower the HDL cholesterol level.

Symptoms of Dyslipidemia

High lipid levels in the blood usually cause no symptoms. Occasionally, when levels are particularly high, lipid is deposited in the skin and tendons. People with high triglyceride levels may develop reddish or yellowish bumps in the skin called eruptive xanthomas. Some people with high cholesterol develop bumps in tendons; bumps in the Achilles tendons in the heel are specific for the inherited disorder familial hypercholesterolemia. Xanthelasma are yellow-white plaques on the eyelids and corners of the eye. Xanthelasma and opaque white or gray rings at the edge of the cornea are sometimes present in people with familial hypercholesterolemia but can also be present in people with normal cholesterol levels.

Very high triglyceride levels can cause the liver or spleen to enlarge, a tingling or burning sensation in the hands and feet, difficulty breathing, and confusion and may increase the risk of developing [pancreatitis](#). Pancreatitis can cause severe abdominal pain and is occasionally fatal.

Symptoms of Dyslipidemia



Achilles Tendon Xanthomas

Achilles tendon xanthomas are diagnostic of familial hypercholesterolemia.

Image courtesy of Michael H. Davidson, MD.



Tendon Xanthomas

Tendon xanthomas are diagnostic of familial hypercholesterolemia.

Image courtesy of Michael H. Davidson, MD.



Eruptive Xanthoma

People who have severe elevations of triglycerides can have eruptive xanthomas over the trunk, back, elbows, buttocks, knees, hands, and feet.

© Springer Science+Business Media



Xanthelasma of the Eyelid

Xanthelasma are yellow-white plaques on the eyelids and corners of the eye. Xanthelasma are sometimes present in patients with familial hypercholesterolemia but can also be present in people with normal cholesterol levels.

Image courtesy of Michael H. Davidson, MD.

Xanthelasma (Eyelid)

This man has yellowish skin growths on his upper eyelids, which can be a sign of high cholesterol levels.

© Springer Science+Business Media

Diagnosis of Dyslipidemia

- Blood tests to measure cholesterol levels

Blood Tests to Measure Cholesterol Levels

LAB TEST

Cholesterol Levels



LAB TEST

Triglycerides Test



Levels of total cholesterol, LDL cholesterol, HDL cholesterol, and triglycerides—the lipid profile—are measured in a blood sample.

When lipid levels in the blood are very high, special blood tests are done to identify the specific underlying disorder. Specific disorders include several hereditary disorders (primary dyslipidemias), which produce different lipid abnormalities and have different risks.

Did You Know...

- Margarines made primarily from liquid oil (squeeze or tub margarines) and those that contain plant stanols or sterols, unlike stick margarines, are healthier substitutes for butter for people who need to limit their butter intake.

Screening for Dyslipidemia

The **lipid profile**, is the levels of total cholesterol, triglycerides, LDL cholesterol, and HDL cholesterol. These levels used to be routinely measured after a person fasts for 12 hours, but in most cases non-fasting testing is sufficient. Doctors usually do this test every 5 years starting at age 20 as part of assessing whether the person is at risk of [coronary artery disease](#).

In addition to measuring lipid levels, doctors also check for other risk factors for cardiovascular disease, such as [high blood pressure](#), [diabetes](#), or a family history of high lipid levels.

In children and adolescents, screening with a non-fasting lipid profile is recommended between the ages of 2 and 8 years if the child has risk factors, such as a family member with severe dyslipidemia or one who developed coronary artery disease at a young age. In children with no risk factors, screening with a non-fasting lipid profile is usually done once before the child reaches puberty (usually between age 9 to 11) and once more between the ages of 17 to 21.

Treatment of Dyslipidemia

- Lose weight
- Exercise
- Decrease saturated fats in the diet
- Often lipid-lowering medications

Usually, the best treatment for people is to lose weight if they have obesity, stop smoking if they smoke, decrease the total amount of saturated fat in their diet, increase physical activity, and then, if necessary, take a lipid-lowering medication.

Regular physical activity can help lower triglyceride levels and increase the HDL cholesterol level. The American Heart Association (AHA) recommends 150 minutes of moderate-intensity exercise or 75 minutes of high-intensity exercise weekly, as well as 2 moderate-intensity strength training sessions weekly.

Treating children may be challenging. The American Academy of Pediatrics and the National Heart, Lung, and Blood Institute recommend treatment for some children with high lipid levels. Diet changes are recommended. Lipid-lowering medications may be given to some children with very high lipid levels who do not respond to changes in diet, particularly children with familial hypercholesterolemia.

Lipid-lowering diet

A lipid-lowering diet is rich in vegetables, fruits, whole grains, and lean proteins, and it minimizes processed foods, saturated fats, and trans fats. People with high triglyceride levels also need to avoid consuming large amounts of sugar, refined flour, and starchy foods.

Eating lots of vegetables, fruits, and whole grains is recommended for healthy cholesterol levels. These foods contain fiber and nutrients. Some foods, including oat bran, oatmeal, beans, peas, rice bran, barley, citrus, strawberries, and apples are particularly rich in soluble fiber and can bind fats in the intestine, helping to lower cholesterol levels.

A lipid-lowering diet contains moderate amounts of protein, emphasizing lean protein sources like chicken and fish, and minimizing red meat, which tends to be higher in saturated fat.

The type of fat consumed is important in a lipid-lowering diet (see [Types of Fat](#)). Fats may be saturated, polyunsaturated, or monounsaturated. Saturated fats increase serum cholesterol levels more than other forms of fat. Saturated fat should provide no more than 5 to 6% of total daily calories. Saturated fats are found in larger amounts in red meats, full-fat dairy products such as cream and butter, and oils from coconut and palm products.

Polyunsaturated fats (which include omega-3 fats and omega-6 fats) and monounsaturated fats may help decrease levels of triglycerides and LDL cholesterol in the blood. Polyunsaturated and monounsaturated fats can be found in vegetable oils, nuts, fatty fish, olive oil, and avocados.

Trans fats, a type of unsaturated fat, are primarily found in heavily processed food. They have been shown to increase LDL cholesterol (the "bad" cholesterol) and decrease HDL cholesterol (the "good" cholesterol), and should be avoided.

Calorie balance is an important aspect to a lipid-lowering diet, as maintaining a healthy weight is critical to managing cholesterol levels.

Did You Know...

- Eating oat bran, oatmeal, beans, peas, rice bran, barley, citrus fruits, strawberries, and apple pulp can help lower cholesterol.

TABLE		
Limiting Fat and Cholesterol in the Diet		
Type of Fat	Food Sources	Recommendations
Monounsaturated	Avocado	Choose these foods in moderation
	Canola oil	
	Safflower oil	

Type of Fat	Food Sources	Recommendations
Polyunsaturated	Olive oil	Choose these foods in moderation
	Nuts	
	Some vegetable oils, such as soybean oil and corn oil Fatty fish, such as salmon, mackerel, herring, and trout Some nuts and seeds, such as walnuts and sunflower seeds	
Saturated	Meats and processed meats Nonskim dairy products, such as whole milk, cheese, and butter Artificially hydrogenated vegetable oils Coconut oil	Eat less of these foods (no more than 5 to 7% of total calories)

Lipid-lowering medications

Treatment with lipid-lowering medications depends not only on the lipid levels but also on whether coronary artery disease, diabetes, or other major [risk factors for coronary artery disease](#) are present. For people who have coronary artery disease or diabetes, the risk of heart attack or stroke can be decreased by the use of the lipid-lowering medications called statins. People who have very high cholesterol levels or who have other high risk factors for heart attack or stroke also may benefit from taking lipid-lowering medications.

There are different types of lipid-lowering medications:

- Statins
- Cholesterol absorption inhibitors
- Bile acid binders
- PCSK9 (proprotein convertase subtilisin/kexin type 9) inhibitors
- Fibric acid derivatives
- Supplements of omega-3 fats
- [Niacin](#)
- [Bempedoic acid](#)

Each type lowers lipid levels by a different mechanism. Consequently, the different types of medications have different side effects and may affect lipid levels differently. Even when taking lipid-lowering medications, people should continue to follow a lipid-lowering diet.

Lipid-lowering medications do more than lower lipid levels—they can also prevent coronary artery disease. In addition, statins have been shown to reduce the risk of early death.

People with very high triglyceride levels and who are at risk of pancreatitis may need both diet changes and triglyceride-lowering medications, usually a fibrate or prescription omega-3 fatty acids.

TABLE			
Lipid-Lowering Medications			
Type	Mechanism of Action	Indications	Some Potential Side Effects
Adenosine triphosphate citrate lyase inhibitor			
Bempedoic acid	Lowers LDL cholesterol by decreasing production in the liver	High LDL cholesterol in people with atherosclerotic cardiovascular disease or familial hypercholesterolemia in people with at least 1 abnormal gene (heterozygous)	Cold or flu-like illness Gout flare Pain in back, arms, or legs Stomach pain
Angiotensin-like protein 3 inhibitor			
Evinacumab	Lowers triglycerides, LDL and HDL cholesterol by increasing clearance from the bloodstream	Familial hypercholesterolemia in people who have 2 abnormal genes (are homozygous)	Decreased energy Dizziness Influenza-like illness Nasopharyngitis Nausea Pain in legs or arms
Bile acid binder			
Cholestyramine Colestipol	Bind bile acids in the intestine, causing the acids to	High LDL cholesterol	Abdominal pain

Type	Mechanism of Action	Indications	Some Potential Side Effects
Colesevelam	be excreted rather than used to make bile and causing the liver to remove more LDL cholesterol from the bloodstream to make bile		Binding of some other drugs (reducing their effectiveness) Bloating Constipation Nausea Increase in triglyceride level (especially in people with high triglyceride levels)
Cholesterol absorption inhibitor			
Ezetimibe	Decreases cholesterol absorption in the small intestine	High LDL cholesterol	Few serious side effects Face and lip swelling (very rare) Loose stools Muscle aches (very rare)
Fibric acid derivatives			
Bezafibrate* Ciprofibrate* Fenofibrate Gemfibrozil	Increase the breakdown of lipids and speed the removal of VLDL from the bloodstream May decrease VLDL production by the liver	High triglycerides Dysbetalipoproteinemia Possibly high VLDL cholesterol	Abdominal pain Bloating Diarrhea Gallstones High liver enzyme levels Muscle aches due to inflammation (myositis) Nausea Rash

Type	Mechanism of Action	Indications	Some Potential Side Effects
Microsomal triglyceride transfer protein inhibitor			
Lomitapide	Decreases production of triglyceride in the liver so that less is secreted into the bile	Familial hypercholesterolemia in people who have 2 abnormal genes (are homozygous)	Diarrhea Liver damage
Niacin			
Niacin	Decreases LDL, VLDL, and triglycerides, but mechanism is not well understood	High triglycerides High LDL and VLDL cholesterol Dysbetalipoproteinemia	Digestive upset Flushing Gout High blood sugar level (hyperglycemia) High liver enzyme levels Itching Ulcers
Omega-3 fatty acids			
Omega-3 fatty acids	Lower levels of triglycerides May decrease production of VLDL	High triglycerides	Belching Diarrhea
PCSK9 inhibitors			
Alirocumab	Lowers level of LDL by increasing removal of LDL from the bloodstream by the liver	Familial hypercholesterolemia and for other people at high risk of coronary artery disease	Flu-like symptoms Rarely, high liver enzyme levels Skin reactions at injection sites
Evolocumab	Lowers level of LDL by increasing removal of LDL	Familial hypercholesterolemia and for other people at high risk	Flu-like symptoms Hives

Type	Mechanism of Action	Indications	Some Potential Side Effects
	from the blood stream by the liver	of coronary artery disease	Skin reactions at injection sites
SiRNA targeting PCSK9			
Inclisiran	Lowers level of LDL by increasing removal of LDL from the blood stream by the liver	Familial hypercholesterolemia and for other people at high risk of coronary artery disease	Skin reactions at injection sites Joint pain or stiffness
Statins (HMG-CoA reductase inhibitors)			
			Bloating Constipation (mild) Fatigue Headache
Atorvastatin	Block the synthesis of cholesterol, increasing the removal of LDL from the bloodstream	High LDL cholesterol, triglycerides, or both	Loose stools
Fluvastatin			Rarely, high liver enzyme levels
Lovastatin			Rarely, muscle aches due to inflammation (myositis) or degeneration (rhabdomyolysis)
Pitavastatin			
Pravastatin			
Rosuvastatin			
Simvastatin			
HDL = high density lipoprotein; HMG-CoA = 3-hydroxy-3-methylglutaryl coenzyme A; LDL = low density lipoprotein; PCSK9 = proprotein convertase subtilisin/kexin type 9; SiRNA = small interfering ribonucleic acid; VLDL = very low density lipoprotein.			
* Not available in the United States.			

Procedures to reduce cholesterol

Medical procedures that lower cholesterol levels are reserved for people with very high levels of LDL cholesterol that do not respond to diet and lipid-lowering medications. Such people include those with familial hypercholesterolemia. LDL apheresis is the most commonly done procedure. LDL apheresis is a non-surgical procedure where blood is drawn from the person and the LDL component is separated

from the rest of the blood in a special machine. The blood (minus the LDL component) is then returned to the person.

Treating causes of elevated cholesterol

Any conditions that cause or are risk factors for elevations in cholesterol levels also need to be treated. So people with [diabetes](#) should carefully control their blood glucose levels. Kidney disease, liver disease, and [hypothyroidism](#) are also treated. If a medication is causing the cholesterol elevation, doctors may give the person a lower dose or a different medication instead.

Monitoring treatment

Doctors usually do blood tests 2 to 3 months after treatment has started to determine whether lipid levels are decreasing. Once lipid levels have decreased sufficiently, doctors do blood tests once or twice a year. Doctors aim for specific targets for lipid levels. These targets are lower if other risk factors for cardiovascular disease are present.

Because some lipid-lowering medications can sometimes cause muscle and liver problems, doctors usually do blood tests before the person starts taking medication. Then, if the person develops side effects, initial (baseline) measurements are available for comparison.

More Information

The following English-language resources may be useful. Please note that The Manual is not responsible for the contents of these resources.

[Healthy Children: Cholesterol Levels in Children and Adolescents](#)

[National Heart Lung and Blood Institute: Blood Cholesterol](#)

Drug Information for the Topic



Copyright © 2026 Merck & Co., Inc., Rahway, NJ, USA and its affiliates. All rights reserved.